Discriminative Dictionary Learning With Motion Weber Local Descriptor for Violence Detection

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***Abstract*— Automatic violence detection from video is a hot topic for many video surveillance applications. However, there has been little success in developing an algorithm that can detect violence in surveillance videos with high performance. In this paper, following our recently proposed idea of motion Weber local descriptor (WLD), we make two major improvements and propose a more effective and efficient algorithm for detecting violence from motion images. First, we propose an improved WLD (IWLD) to better depict low-level image appearance information, and then extend the spatial descriptor IWLD by adding a temporal component to capture local motion information and hence form the motion IWLD. Second, we propose a modified sparse-representation-based classification model to both control the reconstruction error of coding coefficients and minimize the classification error. Based on the proposed sparse model, a class-specific dictionary containing dictionary atoms corresponding to the class labels is learned using class labels of training samples. With this learned dictionary, not only the representation residual but also the representation coefficients become discriminative. A classification scheme integrating the modified sparse model is developed to exploit such discriminative information. The experimental results on three benchmark data sets have demonstrated the superior performance of the proposed approach over the state of the arts.**

***Index Terms*— Class-specific dictionary learning (DL), motion improved Weber local descriptor (MoIWLD), sparse representation, violence detection.**

I. Introduction

VIOLENT behavior seriously endangers social and personal security [1]. A violence detector has immediate applicability both in the surveillance domain and for rating online video contents. Currently, millions of video surveillance devices have been used in public places such as streets, prisons, and supermarkets (some sample frames from benchmark data sets are shown in Fig. 1). Visual surveillance systems collect huge amounts of videos, but humans must still review most of the data to extract informative knowledge. Our goal is to automatically recognize violent behaviors without carefully labeling data over large archives.

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Violence detection involves similar techniques to those used in many related computer vision applications, e.g., action recognition, object detection and surveillance [2]–[8]. Compared with action recognition, relatively little research has been found for detecting action or violent contents. Timely detection of violent outbreaks in crowds may mean the difference between life and death. For this practical consideration, in our work, we focus on the challenging work of detecting violence in surveillance videos and aim to develop a system to effectively detect violent behaviors using computer vision techniques.

Up to now, there have been some developmental systems for detecting violence in videos. Earlier attempts on this have characterized violent scenes by integrating cues obtained from both video and audio tracks. For example, Mahadevan *et al*. [6] proposed to recognize violent scenes by detecting flame and blood and capturing the degree of motion and the characteristic sounds of violent events. Cheng *et al*. [9] recognized gunshots, explosions, and car braking in audio. Cristani *et al*. [7] presented a new method for characterizing audio visual events, where separate audio and video signals were processed in a unique fashion. Lin and Wang [10] presented a novel violent shot detection scheme from both audio and video views (motion, flame and explosion, and blood analysis). Later proposals focused on detecting skin and blood in video sequences, requiring either foreground segmentation or the information of skin color, which performance degraded greatly when the color feature was not discriminating enough. For example, Datta *et al*. [11] exploited an accelerated motion vector to detect the fist fighting and kicking, requiring foreground segmentation to extract the precise silhouettes. Clarin *et al*. [12] proposed a novel system to detect skin and blood colored regions in video sequences and checked if these regions had intensified throughout the whole sequence. Based on activity recognition approaches, Hassner *et al*. [13] proposed the violent flow (ViF) descriptor and developed a novel means for efficient crowd violence detection. However, the performance of this method degrades significantly when dealing with crowded scenes. Zhang *et al*. [14] proposed a fast and robust framework (referred to as RVD) for detecting and localizing violence in surveillance scenes, and the experimental results on several benchmark data sets have demonstrated the superiority of this method over the state of the arts in terms of both detection accuracy and processing speed.

In recent studies, some approaches based on spatiotemporal interest points (STIPs) [15] have been proposed for violence detection. In general, after extracting interest points over the frames, the bag-of-words (BoW) approach is used for

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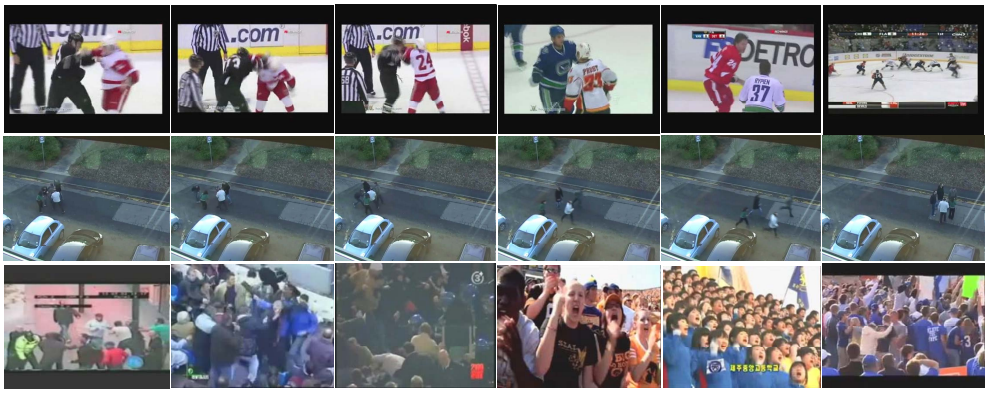

Fig. 1. Sample frames from the Hockey Fight data set (first row), the BEHAVE data set (second row), and the Crowd Violence data set (third row). In each row, the left three columns are violent scenes, while the right three columns are nonviolent scenes.

recognizing violence. These kinds of methods compute only in the regions of interest (located around the detected interest points) and are not discriminative sufficiently. Moreover, the BoW model roughly assigns each feature vector to only one visual word and ignores the spatial relationships among the features. To address these problems, Zhou *et al*. [16] proposed a novel codebook construction method to encode video feature. This approach suits best for structured videos, instead of the videos in our data sets that are more textural.

Targeting on the above challenges, we focus on effective detection of STIPs and feature representation for detecting violent behaviors in real video scenes.

A straightforward way to detect STIPs is to extend the domain of algorithms used to detect 2D interest points [e.g., the widely adopted scale invariant feature transform (SIFT) [17]] to motion domain. Since recognizing human motion is more complicated than recognizing other objects, motion recognition requires enhanced local features that provide both shape and motion information. SIFT is a sparse descriptor that considers the regions of interest only, and are not discriminative enough. To include motion information into an SIFT descriptor, many attempts have been reported [18]. Chen and Hauptmann [19] proposed the motion SIFT (MoSIFT) for interest point detection. MoSIFT not only encodes local appearance of an interest point but also explicitly models the local motion of the point. In the aspect of action recognition, Wang *et al*. [20] have also demonstrated the superiority of the MoSIFT over four existing descriptors, i.e., 3D histogram of gradients (HOG), histogram of optical flow (HOF), HNF (a combination of HOG and HOF), and grid aggregated HOG and HOF.

Another way for violence detection is built on the Weber local descriptor (WLD) [21]. Based on Weber's law, which states that human perception of a pattern depends on not only the change of a stimulus (such as sound, lighting, etc.) but also the original intensity of the stimulus, Chen *et al*. [21] proposed the WLD for face detection. Wang *et al*. [22] further exploited the illumination-insensitive characteristics of the WLD. Li *et al*. [23] proposed a multiscale WLD and a multilevel information fusion approach. Since recognizing violent activities is more complicated than recognizing human faces, violence detection requires enhanced local features that can also provide sufficient motion information. WLD-based interest points with sufficient motion information can provide the necessary information for action recognition. Therefore, motivated by the widely used optical flow approach, which detects the movement of a region by calculating where a region moves by measuring temporal differences, we have recently proposed a novel descriptor, called motion WLD (MoWLD) [24], to measure the spatiotemporal features of dynamic activities effectively. MoWLD can detect a spatially distinctive interest point with a substantial motion. However, we observed from experiments that the original WLD worked well for image areas with rich texture, but gave zero responses for areas that were rather flat. In this paper, we further improve MoWLD by proposing an improved WLD (IWLD) and then forming a motion IWLD (MoIWLD) accordingly.

Once the representation of an action feature is made, action classification can be performed by employing one of the well-known pattern recognition techniques such as nearest neighbor methods based on dimensionality reduction [25], artificial neural networks [26], support vector machines [20], and sparse-representation-based classification (SRC) [27]. Approaches based on local spatiotemporal descriptors are traditionally combined with models of BoW and have achieved promising performance in violence detection [2], [15]. However, the conventional BoW methods rely on the discriminative power of local spatiotemporal descriptors and how often these descriptors occur in the video. Moreover, the performance of a BoW model can be degraded significantly due to a high quantization error. Currently, methods based on sparse coding have been successfully utilized in the areas of action recognition and image classification [18], [28], [29]. The sparse coding methods transform a low-level descriptor to a linear combination of a few atoms in a well-trained dictionary. They usually generate fewer reconstruction errors and can achieve a more discriminative feature representation compared with the BoW models. Wright *et al*. [27] proposed a general classification scheme based on a sparse representation and applied it to face recognition. However, how to learn a discriminative dictionary for both representation and classification of sparse data is still an open problem.

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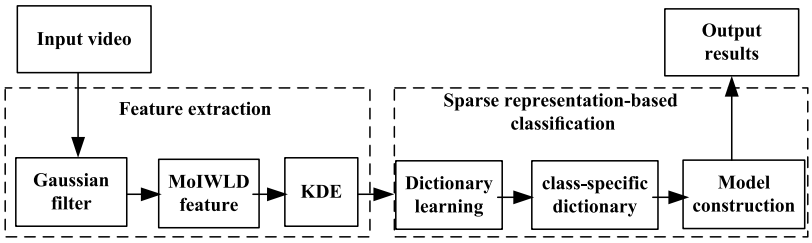

Fig. 2. Framework of the proposed method.

Addressing the above challenges, this paper proposes an effective and robust violence detection algorithm. We propose a novel discriminative holistic descriptor for action representation and a novel supervised learning algorithm based on a modified sparse model to learn a discriminative dictionary for violence classification. We combine the MoIWLD with the sparse coding method in order to generate a more discriminative representation in a video for violence detection. The framework of our approach is illustrated in Fig. 2. First, we extract the MoIWLD features from an input video and employ the kernel density estimation (KDE)-based feature selection method [30], [31] to eliminate redundant and irrelevant features from the extracted MoIWLD. Then, in order to perform violence classification, we propose a modified sparse model to learn a class-specific dictionary, consisting of the dictionary atoms that correspond to the class labels.

Compared with the MoWLD recently proposed in [24], the new contributions presented in this paper are mainly in the following two aspects.

1) In order to detect a sufficient number of interest points containing the necessary information to recognize a violent activity, we follow our recent proposal of MoWLD [24] and present an improved version of MoWLD (i.e., MoIWLD) to extract the low-level image and motion features of a query video more effectively. The proposed MoIWLD detects spatially distinctive interest points having substantial motions. In a sense, this descriptor takes the advantages of both SIFT in terms of computing the histogram using the magnitude and orientation of a gradient and local binary pattern (LBP) [32] in terms of computational efficiency.
2) We propose a modified sparse model that minimizes both the reconstruction error of coding coefficients and the classification error. Based on this model, a class-specific dictionary, i.e., the dictionary atoms that correspond to the class labels, is learned using the class labels of training samples. With this learned dictionary, not only the representation residual but also the representation coefficients become discriminative. A classification scheme integrating the modified sparse model is then developed to exploit such discriminative information.

The experimental results on three challenging benchmark data sets demonstrate the superior performance of our proposed approach over the state of the arts.

The remaining of this paper is organized as follows. Related works on action recognition, anomaly detection, and dictionary learning (DL) are discussed in Section II. Section III introduces our new MoIWLD feature descriptor and the corresponding feature selection method. Section IV details the proposed modified sparse framework and the supervised class-specific DL method for classification. In Section V, the experimental results and analysis are presented. Finally, the conclusions are given in Section VI.

II. Related Works

In this section, we present recent works related to our research and the proposed method on action recognition, anomaly detection, and supervised DL.

A. Related Works on Action Recognition

Violent behavior indicates rapid and significant movement of persons and objects. It is a specific problem within the greater problem of action recognition. We refer readers to [3] and [33]. We classify the most recent research works on action recognition into three groups: statistical, spatiotemporal, and description-based approaches.

Statistical approaches use statistical state-based models to recognize activities. Zhang *et al*. [34] proposed a two-layer hidden Markov model (HMM) framework. This proposed framework is easy to interpret, as both individual actions and group activities have clear meanings. However, the impact of the size of each parameter space for the layered decomposition in this approach was not given. Nguyen *et al*. [35] presented the use of the shared-structure hierarchical HMM to recognize people's behaviors. They have used both the exact approximate inference algorithm and the Rao–Blackwellised particle filter to infer behaviors at different levels. However, it failed to recognize complex behaviors. Shi *et al*. [36] associated the P-Net and the D-condensation algorithm, and provided a natural and efficient way to integrate temporal and logical relationships in daily activities. However, its performance strongly relied on the manually labeled training data. Dai *et al*. [37] introduced an event-based dynamic context model to address the problems of context awareness in the analysis of group interaction scenarios. However, the applicable scenario was very limited. Damen and Hogg [38] presented a novel framework for jointly recognizing and linking visually ambiguous events. However, it failed to recognize complex and ambiguous events.

Spatiotemporal approaches represent activities in volume trajectories as a set of features. These approaches were used to match input with their representative models to determine the activity classes. Bobick and Davis [39] presented a novel

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representation and recognition technique for identifying movements. This approach was based upon temporal templates for dynamic matching in time. However, it was applicable only in those situations where the motion of an object could be easily separated into various simple movements. In [33], an approach based on optical flows was used to represent apparent velocities of movements of brightness patterns in an image, and similar approaches were employed for modeling typical motion patterns [5], [19]. However, this approach would fail in extremely crowded scenes. This issue could be solved by a dense local sampling of optical flows proposed in [40]. Baysal and Duygulu [41] introduced a line-based pose representation and explored its ability in recognizing human actions. They encapsulated a human pose into a collection of line pairs, preserving the geometrical configurations of the components forming the human figure. Saghafi and Rajan [42] proposed a novel embedding that was optimum in the sequence recognition framework based on the spatiotemporal correlation distance as the distance measure. However, its performance mainly relied on the key poses chosen equidistantly from one action period and could not work well in a complex environment. Oikonomopoulos *et al.* [43] represented a human action as a collection of short trajectories extracted in the areas having significant amounts of visual activities in a scene. Vishwakarma *et al.* [44] considered multiclass activities fused in a 3D (spatial and time) coordinate system to achieve maximum accuracy. This method worked well in semantically varying events and was robust to scale and view changes.

Description-based approaches explicitly maintain spatiotemporal structures for human activities. They represent a high-level human activity using the simpler activities, composing the activity and their temporal, spatial, and logical relationships. Yang *et al.* [45] proposed to use a scheme of multifeature learning via hierarchical regression for multimedia semantics understanding. The algorithm could be applied to a wide range of multimedia applications. Gao *et al.* [46] proposed to learn an optimal graph from multiple cues (i.e., partial labels and multiple features) to embed the relationships among data points more precisely. However, this approach would fail in extremely crowded scenes.

B. Related Works on Anomaly Detection

Recently, more and more research attention is given for anomaly detection in video [4], i.e., detecting irregular patterns that are different from regular video events. Although there are many existing works on video anomaly detection [4], [40], few of them can work well in crowded scenes. Mahadevan *et al.* [6] proposed a novel framework for anomaly detection in crowded scenes. Three tasks were designed for anomaly detection. The three tasks are jointly modeling of the appearance and dynamics of a scene, detection of temporal abnormalities, and the detection of spatial abnormalities. Bertini *et al.* [40] constructed a multiscale local descriptor for anomaly detection and achieved real-time performance in video surveillance applications. Roshtkhari and Levine [47] presented a novel approach for video parsing and simultaneous online learning of dominant and anomalous behaviors in surveillance videos. Xu *et al.* [48] presented a novel unsupervised deep learning framework for anomalous event detection in complex video scenes. To exploit the complementary information of both appearance and motion patterns, they introduced a novel double fusion framework, combining both the benefits of traditional early fusion and late fusion strategies.

C. Supervised Dictionary Learning

Sparse representation has received a lot of attention in action recognition area, but most sparse models mainly consider minimizing the reconstruction error, and little attention is paid to better classification. Recent research on supervised DL for sparse coding has been targeted on learning more discriminative sparse models [49]–[51]. According to the predefined relationship between dictionary atoms and class labels, we can divide the existing supervised DL works into three categories: shared DL, class-specific DL, and hybrid DL.

In shared DL, a dictionary shared by all classes is learned, and the discriminative power of the representation coefficients is mined [49], [50], [52]–[54]. Mairal *et al.* [53] proposed a scheme that learned discriminative dictionaries while training a linear classifier over coding coefficients. Based on KSVD [55], Zhang and Li [50] proposed a joint learning framework called discriminative KSVD to learn a dictionary for face recognition. Following [50], Jiang *et al.* [52] proposed to enhance the discriminative power of a learned dictionary via adding a label consistent term. Recently, Mairal *et al.* [49] proposed to learn a dictionary by minimizing different risk functions over the coding coefficients for different tasks, and this learning method is called a task-driven DL. In general, in this scheme, a shared dictionary and a classifier over the representation coefficients are learned together. However, there is no relationship between a dictionary atom and a class label, so no class-specific representation residuals are obtained so that this learning approach is not ideal for classification.

In class-specific DL, a dictionary, in which atoms are predefined to correspond to subject class labels, is learned, and thus the class-specific reconstruction error could be used for classification [51], [56]–[60]. By adding a discriminative reconstruction penalty term in the KSVD model [50], Mairal *et al.* [58] presented a DL algorithm for texture segmentation and scene analysis. Yang *et al.* [57] proposed to learn a structural dictionary by imposing the Fisher discrimination criterion on the sparse coding coefficients to enhance the class discrimination power. By adding nonnegative penalties to both dictionary atoms and representation coefficients, Castrodad and Sapiro [59] proposed to learn a set of action-specific dictionaries. Ramirez *et al.* [56] introduced an incoherence promotion term to the DL model for ensuring the dictionaries representing different classes to be as independent as possible. Wang *et al.* [51] presented a modified sparse framework to learn a dictionary by minimizing the similarity constraint term and the dictionary incoherence term.

Hybrid DL that combines shared DL and class-specific DL has been proposed. Zhou *et al.* [61] proposed to learn a hybrid dictionary using a Fisher-like penalty term on the coding coefficients. Kong and Wang [62] proposed to learn a hybrid

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dictionary by introducing coherence penalty terms on different subdictionaries. Shen *et al*. [63] proposed to learn a dictionary with a hierarchical category structure instead of a flat category structure. However, how to balance the shared part and class-specific part in a hybrid dictionary is not a trivial task.

III. Feature Extraction and Selection

We aim to develop an effective feature representation method, which can generate a more powerful and robust local descriptor to capture the cues for classifying action types in video sequences. Zhang *et al.* [24] proposed a novel MoWLD to extract the low-level features of a query video in spatiotemporal domains. Then, a KDE [24] process was performed in order to eliminate the redundant and irrelevant features. The original WLD with motion information has demonstrated superior performance on action recognition. However, it produces zero response for relatively flat areas.

In this section, for integrity purposes, we first introduce the original WLD, and then propose a more discriminative IWLD. Finally, similar to the work to construct MoWLD shown in [24], details of constructing MoIWLD are presented.

A. Original WLD

Weber's law indicates a fact that, for a stimulus, the ratio between the smallest perceptual change and the background is a constant, and it implies that stimuli are not perceived in absolute terms but in relative terms. In another word, only if the ratio of the change of a stimulus to the original stimulus is big enough, this change can be recognized. Most of us can easily catch a whispered voice in a quiet room, but in a noisy condition, we may not notice someone shouting in our ear. This is the essence of Weber's law.

Inspired by Weber's law, Chen *et al*. [21] proposed a local image descriptor named WLD for the task of face recognition. The descriptor consists of two components, i.e., differential excitation (magnitude) and orientation, which are defined in (1) and (2) [21], [22]. In this paper, we use x_c to denote the central pixel (and also the intensity of the central pixel when there is no confusion) and x_i $(i = 0, 1, \ldots, p-1)$ to represent the p neighboring pixels of x_c (and also their intensities when there is no confusion).

WLD magnitude ξ_m is

$$\xi_m(x_c) = \arctan\left(\alpha \sum_{i=0}^{p-1} \frac{x_i - x_c}{x_c}\right) \tag{1}$$

where the arctangent function is used to prevent the output from being too large and thus partially suppressing the side effect of noise and α is a parameter used to adjust the intensity difference between neighboring pixels.

WLD orientation ξ_o is

$$\xi_o(x_c) = \arctan\left(\frac{x_1 - x_5}{x_3 - x_7}\right) \tag{2}$$

where $x_1 - x_5$ and $x_3 - x_7$ indicate the intensity difference of two neighboring pixels of x_c in the vertical and horizontal directions, respectively.

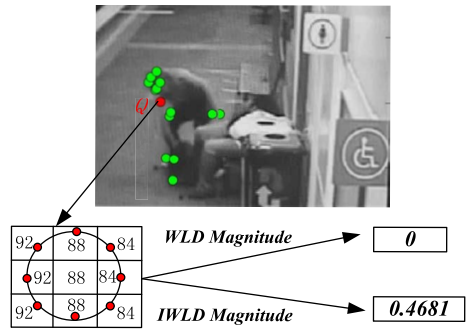

Fig. 3. WLD and IWLD response of a point Q within a nonflat area.

According to [21], ξ_m and ξ_o are then linearly quantized into T (in our experiments, T is set to 12) dominant differential magnitudes and orientations, respectively.

WLD first computes the salient micropatterns by differential excitation, and then builds statistics on these salient patterns along the gradient orientation of the current pixel. In this way, the salient variations within an image can be found to simulate the perception pattern of human beings. Both differential excitation and differential orientation have been proved to be illumination insensitive and computationally efficient [22]. A 2D concatenated histogram based on the differential excitation and orientation can then be constructed to represent each image. As shown in [21] and [23], each row of the 2D WLD histogram corresponds to a dominant differential excitation $\xi_m(x_c)$, and each column corresponds to a dominant orientation $\xi_o(x_c)$. The original WLD histogram [21]–[23] denotes the frequency of a certain dominant differential excitation on a certain dominant orientation.

As seen from its definition, WLD is a kind of dense descriptor that is computed for every pixel and depends on the local intensity variation and the central pixel's intensity. This descriptor employs the advantages of both SIFT in terms of computing the histogram using the gradient and its orientation, and LBP in terms of computational efficiency and smaller support regions. Different from SIFT and LBP, WLD depends on both the local intensity variation and the magnitude of the center pixel's intensity. Since WLD is computed around a relatively small region (e.g., 3×3), while SIFT is computed around a relatively large region (e.g., 16×16), the description granularity of WLD is much smaller than that of SIFT. That is to say, WLD is computed in a finer granularity than SIFT. The smaller size of the support regions for WLD enables WLD to capture more local salient patterns. LBP cannot describe variation extent to the center pixel by its differential excitation, while WLD reflects definite orientation information by the statistic of gradient orientations in local regions.

B. Improved WLD

All of the methods described in [21]–[23] utilized (1) to compute Weber magnitudes and neglected the varying orientations of eight differences. Moreover, when encountered a point on a rather flat area, the ξ_m value in (1) may become zero, although the area may contain certain levels of textures (see Fig. 3).

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In order to address this problem, followed by our previous work [66], we propose an IWLD described as follows.

Denote $X = ((x_0 - x_c/x_c), (x_1 - x_c/x_c), \ldots, (x_{p-1} - x_c/x_c))^T$ and $I = (1, 1, \ldots, 1)^T$. The WLD magnitude in (1) can be rewritten as

$$\xi_m(x_c) = \arctan(\alpha\langle X, I\rangle) \tag{3}$$

where $\langle, \rangle$ is the inner product operator.

Let us denote the angle between x direction and $x_i - x_c$ (i.e., the vector formed by the subtraction of the central pixel's coordinates from the coordinates of the ith neighbor of x_c) by θ_i, for $i = 0, 1, \ldots, p-1$. In our proposed IWLD, xi is the circular neighborhood. Then, we can use the projections of X's components to the x and y directions to compute the effective contributions of the components for the definitions of IWLD magnitudes in the x and y directions, in (4) and (5), respectively.

IWLD magnitude in the x direction is

$$\bar{\xi}_{m_x}(x_c) = \arctan(\alpha\langle X, J_x\rangle) \tag{4}$$

where $J_x = (\cos\theta_0, \cos\theta_1, \ldots, \cos\theta_{p-1})^T$.

IWLD magnitude in the y direction is

$$\bar{\xi}_{m_y}(x_c) = \arctan(\alpha\langle X, J_y\rangle) \tag{5}$$

where $J_y = (\sin\theta_0, \sin\theta_1, \ldots, \sin\theta_{p-1})^T$.

Then, we define the IWLD magnitude as

$$\bar{\xi}_m(x_c) = \sqrt{(\arctan\alpha\langle X, J_x\rangle)^2 + (\arctan\alpha\langle X, J_y\rangle)^2} \tag{6}$$

and the IWLD orientation is defined as

$$\bar{\xi}_o(x_c) = \arctan\left(\frac{\bar{\xi}_{m_y}(x_c)}{\bar{\xi}_{m_x}(x_c)}\right). \tag{7}$$

Comparing the IWLD with the original WLD, we can see that the IWLD represents local patterns more effectively and accurately in the patch. As shown in Fig. 3, when a point falls in a nonflat area, (6) gives a nonzero response.

C. IWLD Histogram

Collecting the values of Weber magnitudes and orientations in IWLD on an image region, we can build an IWLD histogram for the region. However, the resultant IWLD histogram is not rotation invariant, and is sensitive to partial occlusion and deformation. Aiming to address these two problems, we propose to rebuild the IWLD histogram by aggregating the IWLD histograms of neighboring regions and also aligning these histograms to their dominant orientation, as follows.

1) The IWLD magnitude and orientation are calculated according to (6) and (7) for every pixel in a region of a Gaussian-blurred image F.
2) The IWLD orientation is quantified into 12 dominant bins using the nonlinear quantization method in [23], with each bin covering 30°. An orientation histogram with 12 bins is then formed.
3) An aggregated histogram of IWLD magnitudes and orientations from all neighboring regions is formed as the feature representation of a local appearance. This representation makes the IWLD be tolerant to partial occlusion and deformation.
4) When an interest point is detected, the corresponding dominant orientation can be determined. The locations of all IWLD magnitudes in the neighboring regions are then rotated according to the dominant orientation to achieve the rotation invariance.
5) Pixels in the neighboring region are normalized into 144 (16×9) elements, which are grouped into 16 (4×4) grids of 9 pixels each, as shown in Fig. 4. Each grid has its own IWLD orientation histogram describing the orientation of the subregion. This results in an IWLD feature vector of 192 dimensions ($4 \times 4 \times 12 = 192$).

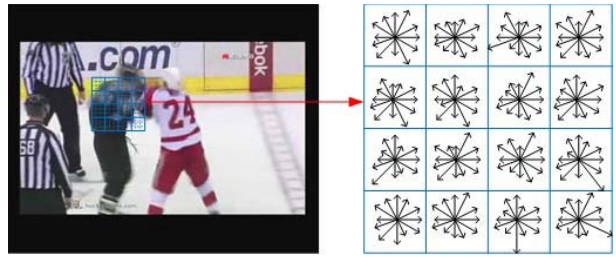

Fig. 4. Grid aggregation for IWLD feature descriptors. Pixels in a neighborhood are grouped into 4×4 grids, each containing $3 \times 3 = 9$ pixels. An orientation histogram with 12 bins is formed for each grid resulting into a 192-element vector for the neighborhood.

Fig. 4 illustrates the idea of the IWLD histogram grid aggregation. Pixels in a neighborhood are grouped into 4×4 blocks, each containing 3×3 pixels. By constructing the IWLD feature vector in this way, we can obtain a more discriminative descriptor of $16 \times 12 = 192$ dimensions in total.

D. Motion IWLD

The IWLD feature descriptor describes only the properties of still images and carries no motion information of a video. Therefore, the detected candidate points are distinctive in appearance only, but are independent of the motions or actions in a video. Clearly, motion information is essential for extracting interest points to eliminate irrelevant information for action recognition. In our previously proposed MoWLD algorithm [24], we adopted the widely used optical flow approach to detect the movement within an image region. A local extreme among all IWLD feature points can become only an interest point if it has sufficient motion in optical flow field.

The optical flow approach detects the movement of a region by calculating the temporal differences of the region in two consecutive frames. Compared with video cuboids or volumes, optical flow explicitly captures the magnitude and direction of a motion, instead of implicitly modeling motion through appearance changes over a time period. Explicitly measuring motion is beneficial for recognizing actions. In our work, to add motion information into the IWLD, we apply the same aggregation idea as that for generating the IWLD to integrate the optical flow of every grid in a region into the IWLD and form the MoIWLD.

Surveillance videos are typically captured by stationary cameras, so the direction of a movement generated by violent

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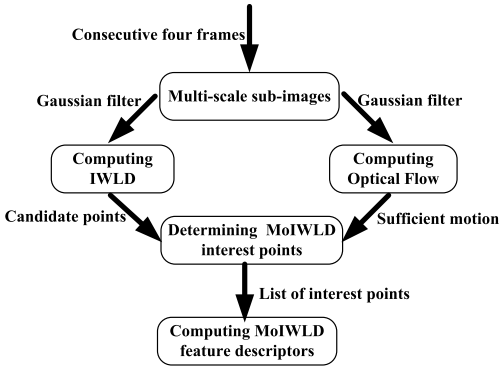

Fig. 5. Flowchart of our MoIWLD algorithm. Four consecutive frames are input to compute the IWLD and optical flow. Candidate points with sufficient motion are determined as the MoIWLD interest points, for which MoIWLD features are extracted.

actions is typically irregular and variable, and it can be used to distinguish the violent actions from normal actions. The dominant orientation feature is a main difference between IWLD and an optical flow. The magnitude and direction of a movement can be used to construct an optical flow histogram similar to the construction of an IWLD histogram. The orientations of optical flows in each grid are normalized into 12 directions, and an optical flow histogram of 12 bins is constructed for each grid. For a 4×4 grid neighborhood, this results in an aggregated optical flow histogram of a dimension of $4 \times 4 \times 12 = 192$.

Furthermore, to create a more robust descriptor that contains both temporal and contextual information, we also integrate the IWLD and optical flow histograms of three previous frames into the descriptor. Therefore, all four sets of the aggregated IWLD and optical flow histograms on each Gaussian filtered image are concatenated to form the MoIWLD (as shown in Fig. 5), which now has 1536 ($4 \times 2 \times 192 = 1536$) dimensions.

E. Multiscale MoIWLD

The IWLD features described above are extracted from small patches (i.e., grids) of the same size (i.e., sclae) of 3×3 pixels. However, subimages containing various human structures may have different sizes. Thus, we conduct a multiscale image analysis to extract more discriminative and robust features of different human local structures. We adopt the multiscale feature analysis approach [21] to develop a multiscale IWLD for each image on a Gaussian pyramid (see Fig. 5). The multiscale IWLD provides local salient patterns around a point (e.g., the central pixel in Fig. 6) at various scales. It is computed on a set of circular symmetric neighborhoods consisting of various numbers of pixels, as shown in Fig. 6.

Furthermore, multiscale optical flows are also computed on each Gaussian filtered image. Multiple-scale optical flows are calculated according to the IWLD scales. A local extreme of the multiscale IWLD feature points can only become an interest point if it also has sufficient motion in terms of multiscale optical flows in the Gaussian pyramid. We assume that a

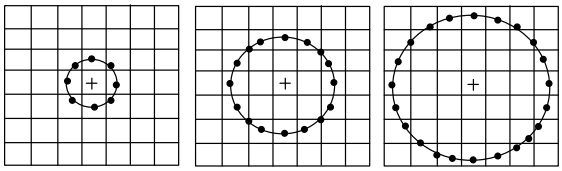

Fig. 6. Set of circular symmetric neighborhoods consisting of various numbers of pixels.

complicated action can be represented by the combination of a reasonable number of interest points. Therefore, we do not assign strong constraints to STIPs. As long as the candidate interest points have made movements with distances larger than a minimum value, they can be deemed as MoIWLD interest points. The extracted MoIWLD interest points are invariant to scale and rotation in the spatial domain, but they are not invariant to scale in the temporal domain. The multiscale MoIWLD selects distinctive interest points with sufficient motions from which humans can see the actions happened at the corresponding points and machines can learn an action model.

Since the multiscale MoIWLD is based on IWLD and optical flow, it has the advantages described as follows. Instead of combining the histograms obtained from IWLD and optical flows, we build a single feature descriptor through the early fusion that concatenates both histograms into one vector. This single future descriptor captures, at the same time, the appearance and motion information that are essential for classifying actions. Moreover, the MoIWLD captures local appearance using an aggregated HOG in neighboring regions, so it is tolerant to partial occlusion and deformation. Furthermore, when an interest point is detected, a dominant orientation is calculated and all gradients in the neighborhood are rotated according to the dominant orientation. Therefore, the multiscale MoIWLD is rotation invariant.

F. KDE-Based Feature Selection

To improve both performance and computational efficiency, we employ a nonparametric density estimation method to select the most representative features from the extracted MoIWLD features.

Nonparametric density estimation, an important tool for statistical analysis of data, is an alternative to parametric approaches, in which one specifies a model up to a small number of parameters and then estimates the parameters via the likelihood principle. Being a classic nonparametric density estimation method, the KDE-based feature selection method [30], [31] depends only on data samples to get an estimation and does not need a prior knowledge of the data distribution. As long as there are enough samples, KDE can give a satisfactory approximation of an underlying distribution no matter whether it is regular or irregular, and single modal or multimodal. Therefore, we employ KDE to find the discriminative features that have multimodal distributions.

Suppose $x_1, x_2, \ldots, x_N$ are N independent and identically distributed data of a 1D random variable x. KDE infers the probability density function (PDF) of x by centering a kernel

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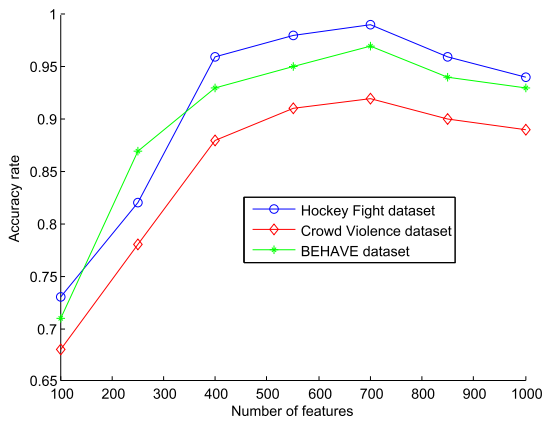

Fig. 7. Variation of the performance with the change of number of features on three benchmark data sets.

function $K(x)$ at each data point x_i as

$$f_h(x) = \frac{1}{hN} \sum_{i=1}^{N} K\left(\frac{x - x_i}{h}\right) \tag{8}$$

where h is a smoothing parameter, named bandwidth, which can be adaptively chosen using the method proposed in [31].

In order to reduce the dimension of the MoIWLD feature space, we use KDE to obtain a smooth PDF based on our training data. However, the common Gaussian kernel density estimator [30] lacks local adaptivity, and this often results in a higher sensitivity to outliers. Therefore, an adaptive kernel, like the one discussed in [31], is chosen to improve local adaptivity and reduce bias.

If the PDF of a feature is bimodal or multimodal, this feature is considered to be more discriminative than those of only a single mode. We estimate the PDF of each feature on the original 1536 MoIWLD features. According to the number of modes, we sort the 1536 MoIWLD features in descending order. Finally, the first 550 features are selected to form the reduced MoIWLD, which is more effective than the original ones (as shown in Fig. 7).

IV. Sparse Representation and Dictionary Learning

Once we have obtained action feature representation, action classification can then be performed by employing pattern recognition technologies. Wright *et al.* [27] proposed a general classification scheme based on sparse representation and applied it for robust face recognition. However, how to learn a discriminative dictionary for both sparse data representation and classification is still a challenging problem.

Note that the conventional SRC scheme does not consider the relationship between the subdictionaries of two classes. It is used to minimize reconstruction error instead of classification error. In addition, the discriminative information in training samples cannot be sufficiently exploited by such a naively supervised DL method.

As an improvement, in our proposed model (detailed as follows), each sample is locally approximated by a linear combination of its nearby samples and a classification error is introduced. The proposed sparse representation model selects the samples that can best represent their own action classes as the bases of their class-specific dictionaries. Specifically, we combine the reconstruction error with a classification error to form a unified objective function. The learned dictionary encourages the signals from the same class to have similar sparse codes and those from different classes to have dissimilar sparse codes to achieve accurate classification results.

A. Original SRC Model

Wright *et al.* [27] proposed a general SRC scheme, in which the training samples of all classes were taken to form the dictionary representing a query face image. The query image was classified by evaluating which class led to the minimal error of reconstructing it.

Given K classes of subjects, let $D = [A_1, A_2, \ldots, A_K]$ be the dictionary formed by A_i, where A_i $(i = 1, 2, \ldots, K)$ is the subset of training samples of class i. Let y be a test sample. The SRC algorithm is summarized as follows.

1) Normalize each training sample $A_i, i = 1, 2, \ldots, K$.
2) Define and solve the l_1-minimization problem: $\hat{x} = \arg\min_x \{\|y - Dx\|_2^2 + \gamma \|x\|_1\}$, where γ is a scalar constant.
3) Label the test sample y by: $\text{Label}(y) = \arg\min_i \{e_i\}$, where $e_i = \|y - A_i \hat{\alpha}^i\|_2^2$, with $\hat{\alpha}^i$ representing the coefficient vector associated with class i.

Obviously, the underlying assumption of this scheme is that a test sample can be represented by a weighted linear combination of just those training samples belonging to the same class. Its impressive performance reported in [27] showed that sparse representation is naturally discriminative.

B. Proposed SRC Model

In our proposed model, two terms, i.e., the representation-constrained term and the coefficient adjustment term, are introduced and described in the following to ensure that the learned dictionary is sufficiently discriminative. The representation-constrained term is utilized to enforce the class-specific subdictionary to have a good capability when reconstructing a query image using training samples having the same class label. On the other hand, the coefficient adjustment term is utilized to enforce the class-specific subdictionary to have a poor capability when reconstructing a query image using training samples with different class labels. Based on these two terms, a classification scheme is then developed to exploit the discriminative information.

The samples in different action classes have different MoIWLDs, so we will focus on class-specific DL. In a class-specific DL, each dictionary atom in the learned dictionary, denoted by $D = [d_1, d_2, \ldots, d_k]$, has a class label corresponding to each subject class. d_i $(i = 1, 2, \ldots, K)$ in D is the subdictionary corresponding to class i. By representing a test sample over the learned dictionary D, the representation residual associated with each class can be employed to classify it, as in the SRC method.

Given training MoIWLD feature samples $\{a_{ij} | i = 1, 2, \ldots, K$ and $j = 1, 2, \ldots, N\}$, where a_{ij} is the jth sample of class i, N denotes the number of training samples

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in each class, and K is the number of classes. Let $A = [A_1, A_2, \ldots, A_i] \in \mathbb{R}^{n\times N}$, where $A_i = [a_{i1}, a_{i2}, \ldots, a_{iN}]$, $i = 1, 2, \ldots, K$, and n is the MoIWLD feature dimension. We aim to include the classification error as a term in the objective function for DL in order to make the dictionary be optimal for classification. The sparse code Z can be directly used as a feature for classification. Here, we use a linear predictive classifier $f(Z_i; W) = WZ_i$, where $Z = [Z_1, Z_2, \ldots, Z_i]$. Denote the learned dictionary by $D = [d_1, d_2, \ldots, d_k] \in \mathbb{R}^{n\times k}$ ($k > n$ and $k \ll N$). We propose the following modified sparse model:

$$\begin{aligned}\langle D, W, Z\rangle = \arg\min_{D,W,Z} &\left\{\|A - DZ\|_F^2 + \lambda_1\|Z\|_1 + \lambda_2\|Z - m\|_F^2\right.\\ &\left.+ \gamma_1\|WZ - B\|_F^2 + \gamma_2\|W\|_F^2\right\}\\ \text{s.t. } &\|d_c\|_2 \le 1 \quad \forall c \in \{1, 2, \ldots, k\}\end{aligned} \tag{9}$$

where $Z = [Z_1, Z_2, \ldots, Z_i] \in \mathbb{R}^{k\times N}$ is the matrix consisting of the coding coefficients of A_i over D, $m = [m_1, m_2, \ldots, m_i] \in \mathbb{R}^{k\times N}$, m_i denotes the mean vector of Z_i in class i, $\|WZ - B\|_F^2$ represents the classification error, and $B = [0, 0 \ldots, b_N] \in \mathbb{R}^{m\times N}$ are the class labels of input signals A_i. $b(i) = [0, 0 \ldots 1 \ldots 0, 0]^T \in \mathbb{R}^m$ is a label vector corresponding to an input signal A_i, where the nonzero position indicates the class of A_i and $\|\cdot\|_F$ denotes Frobenius norm. $W \in \mathbb{R}^{m\times k}$ denotes the matrix of classifier parameters, and λ_1, λ_2, γ_1, and γ_2 are the scalars controlling the relative contributions of the corresponding terms.

The first two terms on the right-hand side of (9) can be seen as the basic model of class-specific DL. Different from the conventional sparse model SRC in [27], in our model, the representation-constrained term $\phi = \lambda_2\|Z - m\|_F^2 + \gamma_1\|WZ - B\|_F^2$ and coefficient incoherence term $\psi = \gamma_2\|W\|_F^2$ are introduced in (9).

1) Representation-Constrained Term: Z_i denotes the sparse coefficients of A_i over dictionary D, so $A_i \approx DZ_i$. Since Z_i is associated with class i, it is naturally expected that Z_i can be well represented by only m_i because m_i denotes the mean vector of Z_i in class i. Therefore, there should exist a Z_i such that $\|Z_i - m_i\|_F^2$ is small. This term can control the reconstruction error of coefficients Z_i. On the other hand, W denotes the matrix of classifier parameters and B records the class labels, so $\gamma_1\|WZ - B\|_F^2$ represents the classification error, and minimizing this term is to minimize the classification error.

Overall, minimizing the representation-constrained term defined by $\phi = \lambda_2\|Z - m\|_F^2 + \gamma_1\|WZ - B\|_F^2$ minimizes both the reconstruction error and classification error.

2) Coefficient Adjustment Term: Given a test sample, Wright *et al*. [27] suggested that its sparse representation could be found by an SRC scheme, and in the sparse coefficients recovered by SRC, the largest coefficients were associated with the training samples that had the same class label as the test sample. It implies that the test sample can be approximated by a weighted linear combination of its own training samples with these largest coefficients. Likewise, in our proposed class-specific DL, because W denotes the set of classifier parameters, minimizing $\|W\|_F^2$ is for the class-specific DL to reach the minimum classification error.

On the other hand, minimizing the coefficient adjustment term and representation-constrained term is efficient for classification, which allows feature sharing among the classes. We will show that good classification results can be obtained using only a single unified dictionary by a simple extension to the objective function for joint dictionary and classifier construction. It encourages that the largest classification parameters of training samples from a different class over D are associated with the corresponding different subdictionary. Therefore, the coefficient adjustment term enforces a label consistency constraint on the sparse codes.

C. Supervised Class-Specific Dictionary Learning

Although the objective function in (9) is not jointly convex to (D, W, Z), it is convex with respect to each of D, W, and Z when the other two parameters are fixed. Therefore, (9) can be divided into three subproblems by optimizing D, W, and Z, respectively, i.e., updating Z while fixing D and W, updating D while fixing W and Z, and updating W while fixing D and Z, detailed as follows.

1) Updating Z: When D and W are fixed, the objective function in (9) can be regarded as sparse coding problem for solving $Z = [Z_1, Z_2, \ldots, Z_K]$. When Z_i is updated, all $Z_j (j \neq i)$ are also fixed. Thus, for each Z_i, the objective function in (9) can be replaced by

$$\begin{aligned}&\|A - DZ\|_F^2 + \lambda_1\|Z\|_1 + \lambda_2\|Z - m\|_F^2\\ &\quad + \gamma_1\|WZ - B\|_F^2\end{aligned} \tag{10}$$

and (10) can be rewritten as

$$\begin{aligned}\langle Z_i\rangle = \arg\min_{Z_i} &\left\{\|A_i - DZ_i\|_2^2 + \lambda_1\|Z_i\|_2^2\right.\\ &\left.+ \lambda_2\|Z_i - m_i\|_2^2 + \gamma_1\|WZ_i - b_i\|_2^2\right\}.\end{aligned} \tag{11}$$

By solving (11), we have

$$\begin{aligned}Z_i = &\{D^T D + (\lambda_1 + \lambda_2)I + \gamma_1 W^T W\}^{-1}\\ &\times (D^T A_i + \lambda_2 m_i + \gamma_1 W^T b_i).\end{aligned} \tag{12}$$

2) Updating D: When Z and W are fixed, (9) can be regarded as solving $D = [D_1, D_2, \ldots, D_K]$ sparse coding problem. When D_i is updated, all $D_j (j \neq i)$ are fixed. Thus, (9) can be replaced by

$$\begin{aligned}\langle D\rangle = &\arg\min_{D} \|A - DZ\|_F^2\\ &\text{s.t. } \|d_c\|_2 = 1 \quad \forall c \in \{1, 2, \ldots, k\}.\end{aligned} \tag{13}$$

The subproblem in (13) can be solved effectively by the Lagrange dual method [64].

3) Updating W: When D and Z are fixed, (9) can be replaced by

$$\arg\min_{W} \left\{\gamma_1\|WZ - B\|_F^2 + \gamma_2\|W\|_F^2\right\}. \tag{14}$$

Letting $W = [W_1, W_2, \ldots, W_i]$, (14) can be rewritten as

$$\arg\min_{w_i} \left\{\gamma_1\left(\|W_i Z_i - b_i\|_2^2 + \frac{\gamma_2}{\gamma_1}\|W_i\|_2^2\right)\right\}. \tag{15}$$

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In (15), $\|W_i Z_i - b_i\|_2^2 + (\gamma_2/\gamma_1)\|W_i\|_2^2$ can be further rewritten as

$$\|W_i Z_i - b_i\|_2^2 + \rho\|W_i\|_2^2. \tag{16}$$

Denote $\rho = \frac{\gamma_2}{\gamma_1}$. Obviously, (16) can be solved using the least square method. Thus, we can get the following solution:

$$W_i = b_i Z_i^T \left(Z_i Z_i^T + \frac{\gamma_2}{\gamma_1} I\right)^{-1}. \tag{17}$$

Thus, based on the above equations, we can get the optimized values of all parameters for (9).

D. Classification Scheme

Once the dictionary D has been learned, it can be adopted to represent a testing sample y and perform classification.

We propose the following representation model:

$$\hat{\alpha} = \arg\min_{\alpha} \left\{\|y - D\alpha\|_F^2 + \gamma\,\|\alpha\|_2\right\} \tag{18}$$

where γ is a constant value, and $\hat{\alpha} = [\hat{\alpha}^1, \hat{\alpha}^2, \ldots, \hat{\alpha}^K]^T$, where $\hat{\alpha}^i$ is the subvector associated with subdictionary D_i. In the learning stage, we have enforced the class-specific representation residual to be discriminative. Therefore, if y is from class i, the residual $\|y - D_i\hat{\alpha}^i\|_2^2$ should be very small; otherwise, $\|y - D_j\hat{\alpha}^j\|_2^2$, $j \neq i$ should be big. In addition, the coefficient vector $\hat{\alpha}$ should be far different from the coefficient vector of other classes. Based on the discrimination capability of both representation residual and coefficient vector, the metric for classification can be defined as

$$l = W\hat{\alpha}. \tag{19}$$

Finally, we simply use the linear predictive classifier to estimate the label i of vector l, and the label index corresponds to the largest element of the vector l.

V. Experiments and Results

To evaluate the performance of our proposed ideas, we compare our method against the state-of-the-art approaches either implemented by us or cited from the literature, including the BoW-based methods, the RVD violence detection method in [14], the appearance and motion DeepNet (AMDN) method in [48], the ViF method in [13], the method in [27], and our recently published method in [24]. To evaluate the classification accuracy, we employ the fivefold cross-validation test on each data set. The results are reported with mean prediction accuracy (ACC)± standard deviation (SD), as well as the area under the ROC curve (AUC). Furthermore, in our proposed model, there are two stages, i.e., DL stage and classification stage. In the DL stage, we set $\lambda_1 = 0.005$, $\lambda_2 = 3$, $\gamma_1 = 1$, and $\gamma_2 = 0.1$, and in the classification stage, we set $\gamma = 0.01$. Next, we first briefly introduce the three benchmark data sets, and then present the experimental results with discussion.

A. Data Sets

Experiments of our method were conducted on three challenging benchmark data sets, i.e., the Hockey Fight data set [13], the BEHAVE data set [65], and the Crowd Violence data set [13].

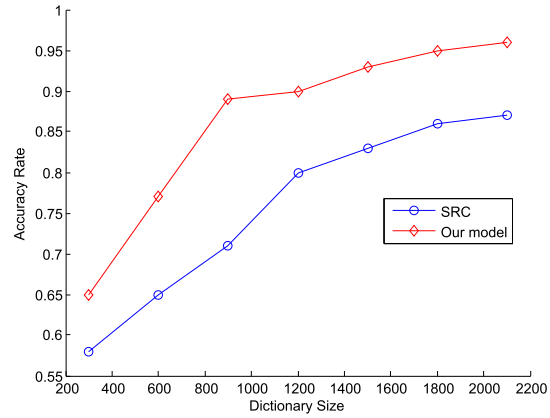

Fig. 8. Performance comparison of our sparse classification model with the original SRC classification model on benchmark data sets.

TABLE I
Comparison on Average Classification Time (in Milliseconds) on the Three Benchmarking Data Sets

Method	Hockey Fight	BEHAVE	Crowd Violence
SRC	100.19	190.81	94.49
Proposed method	4.67	8.38	4.15

The Hockey Fight data set contains 1000 video clips of actions from hockey games of the National Hockey League, of which 500 are manually labeled as fight and others are labeled as nonfight. Each clip consists of 50 frames (with a resolution of 360 × 288 pixels).

The BEHAVE data set contains more than 200 000 frames (with a resolution of 640 × 480 pixels) and various scenarios, including walking, running, chasing, discussing in groups, driving or cycling across the scene, fighting, and so on. We partition the data set into clips with various activities and manually label them as violence or nonviolence. Each clip consists of at least 100 frames. Finally, we pick 80 clips for violence detection, including 20 violence clips and 60 nonviolence clips.

The Crowd Violence data set is assembled for testing violent crowd behavior detection. All video clips are collected from YouTube, presenting a wide range of scene types, video qualities, and surveillance scenarios. The data set consists of 246 video clips including 123 violent clips and 123 normal clips with a resolution of 320 × 240 pixels. The whole data set is split into five sets for fivefold cross validation. Half of the footages in each set present violent crowd behaviors and the other half present nonviolent crowd behaviors.

B. Results and Discussion

1) Proposed SRC Versus the Original SRC: In order to demonstrate the performance of the proposed sparse classification model proposed in Section IV, we first compare our method with the original SRC classification algorithm on the three databases. The average results shown in Fig. 8 demonstrate that our proposed model achieves a higher classification rate than the original SRC algorithm.

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TABLE II
ACCURACY COMPARISON OF VARIOUS FEATURE REPRESENTATIONS PAIRED WITH BOW FOR VIOLENCE DETECTION ON THE HOCKEY FIGHT DATA SET

Vocabulary	HOG+BoW [2]	HOF+BoW [2]	MoSIFT+BoW [2]	MoWLD+BoW	MoIWLD+BoW
300 words	90.8%	87.2%	90.4%	91.3%	92.2
600 words	91.4%	87.4%	90.5%	91.7%	92.8
900 words	91.6%	88.5%	90.9%	91.9%	93.2
1200 words	91.4%	88.3%	90.6%	91.8%	93.7
1500 words	91.2%	88.2%	90.6%	91.5%	93.9
1800 words	91.2%	88.4%	90.5%	91.1%	94.1
2100 words	90.7%	87.9%	90.4%	91.2%	94.3

TABLE III
DETECTION RESULTS ON THE HOCKEY FIGHT DATA SET

Algorithm	ACC±SD	AUC
HOG+BoW [2]	88.77±0.73%	0.9123
HOF+BoW [2]	86.07±0.59%	0.8843
HNF+BoW [2]	89.27±0.79%	0.9294
ViF [13]	90.07±0.99%	0.9429
MoSIFT+BoW [2]	88.8±0.75%	0.9052
MoWLD+BoW	89.28±0.93%	0.9112
MoIWLD+BoW	91.8±1.03%	0.9412
RVD [14]	92.1±1.01%	0.9496
AMDN [48]	89.7±1.13%	0.9198
SRC [27]	94.4±1.07%	0.9623
MoWLD+Sparse Coding [24]	93.8±1.08%	0.9618
Proposed method	96.8±1.04%	0.9808

TABLE IV
DETECTION RESULTS ON THE BEHAVE DATA SET

Algorithm	ACC±SD	AUC
HOG+BoW [2]	58.97±0.34%	0.6394
HOF+BoW [2]	60.03±0.28%	0.5923
HNF+BoW [2]	58.24±0.31%	0.6113
ViF [2]	83.62±0.19%	0.8632
MoSIFT+BoW [2]	62.78±0.23%	0.6679
MoWLD+BoW	81.65±0.18%	0.8324
MoIWLD+BoW	81.98±0.15%	0.8415
RVD [14]	85.29±0.16%	0.8878
AMDN [48]	84.22±0.17%	0.8562
SRC [27]	82.7±0.14%	0.8538
MoWLD+Sparse Coding [24]	87.07±0.13%	0.8928
Proposed method	88.83±0.11%	0.9108

Also, the comparison result on their average computation times for classifying images is shown in Table I. As our algorithm learns a single overcomplete dictionary and an optimal linear classifier jointly, compared with the SRC model, on average, it is around 25 times more efficient.

2) Results on the Hockey Fight Data Set: We empirically demonstrate the superior performance of our ideas of MoWLD and MoIWLD over existing popular feature descriptors, including HOG, HOF, and MoSIFT [2]. Table II shows the results on the Hockey Fight data set, where HOG + BoW, HOF + BoW, and MoSIFT + BoW refer to HOG, HOF, and MoSIFT when paired with the BoW model.

As shown in Table II, MoSIFT and HOG-based BoW models perform comparably, with slight better results achieved by HOG compared with MoSIFT. Our recently proposed MoWLD (noted as MoWLD + BoW in Table II) outperforms all the above-mentioned approaches. MoIWLD has further improved the results, so MoIWLD is more discriminative and effective.

Furthermore, to assess the impact of dictionary size on classification accuracy, we have run the experiments with dictionaries of different sizes. As shown in Table II, with the increase in the dictionary size, the performance begins to rise and then stays stable. This phenomenon indicates that selecting an appropriate dictionary size is significant to both accuracy and computational efficiency.

Table III shows the results obtained by the proposed approach after adopting the BoW approach and the proposed sparse classification model into the MoIWLD approach. The dictionary size is fixed to 1800 in this experiment. We compare the results with not only HOG, HOF, HNF, MoSIFT, MoWLD, and MoIWLD paired with BoW, but also ViF, RVD, and AMDN. The RVD method [14] adopts a Gaussian model of optical flow to extract candidate violence regions, which has reduced many noise disturbances, so its performance is better than the BoW-based approaches. The AMDN approach [48] utilizes optical flow as the input image feature, and there exist many redundant and interference features, so its performance is not very good. It can be seen from Table III that using our proposed MoIWLD together with the proposed sparse classification model has performed the best. Furthermore, our approach outperforms the competing SRC algorithm and MoWLD + sparse coding with the same size of dictionary. This is due to the fact that our proposed supervised class-specific DL framework incorporates representation-constrained and coefficient adjustment terms resulting in the highest recognition rate.

3) Results on the BEHAVE Data Set: We compare our approaches with the state-of-the-art approaches implemented by us on the BEHAVE data set, where 20 clips of this data set are randomly picked for training. Table IV presents the results obtained with the above-mentioned methods on this data set. The dictionary size is fixed to 1800 in this set of experiments. As it can be seen from Table IV, our proposed SRC-based method outperforms other approaches. This again demonstrates that the proposed approach is significantly superior in performance to all other approaches. The performance of the RVD method [14] and the AMDN [48] on this data set is consistent with their performance on the Hockey Fight data set. Furthermore, our MoIWLD combined with our proposed sparse classification method outperforms SRC methods. It validates that the representation-constrained term and coefficient adjustment term can improve the discriminative

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Fig. 9. Examples of false alarms on the BEHAVE data set.

Fig. 10. Examples of false alarms on the Crowd Violence data set.

TABLE V
DETECTION RESULTS ON THE CROWD VIOLENCE DATA SET

Algorithm	ACC±SD	AUC
HOG+BoW [2]	57.98±0.37%	0.6252
HOF+BoW [2]	58.71±0.12%	0.5931
HNF+BoW [2]	57.05±0.32%	0.6154
ViF [2]	82.13±0.21%	0.8595
MoSIFT+BoW [2]	57.09±0.37%	0.6073
MoWLD+BoW	88.16±0.19%	0.9028
MoIWLD+BoW	88.78±0.19%	0.9109
RVD [14]	82.89±0.19%	0.8559
AMDN [48]	84.72±0.17%	0.8891
SRC [27]	89.6±0.18%	0.9288
MoWLD+Sparse Coding [24]	89.38±0.13%	0.9398
Proposed method	93.19±0.12%	0.9508

ability of sparse representation model. The results on this data set demonstrate that our algorithm is also effective for detecting violence in a group fighting scene. False alarms happen only when a group of people get together to do some strenuous nonviolence activities (example frames are shown in Fig. 9).

4) Results on the Crowd Violence Data Set: This data set is more challenging than the other two data sets because it contains many crowded scenes. The set contains 246 clips divided into five splits, each containing 123 violent and 123 nonviolent scenes. Table V presents the results obtained using different methods mentioned above on this data set. The dictionary size is again fixed to 1800 in this set of experiments. In this data set, due to more crowded scenes, the detection rate of RVD method decreases. On the contrary, the performance of AMDN method is still very stable. However, because of the introduction of optical flow noise, AMDN's performance is not very good. Our proposed sparse classification-based approach still outperforms other approaches. MoIWLD is still significantly superior in performance to HOG, HOF, HNF, RVD, AMDN, and MoWLD. It confirms that our proposed MoIWLD is a more effective descriptor for describing action feature. Consistent with the results on the previous two data sets, our MoIWLD combined with the proposed sparse classification method outperforms the SRC methods. It indicates that the proposed classification model has a smaller classification error rate compared with the original SRC. The results on this data set demonstrate that our algorithm is also effective for detecting violence in a crowded scene. Some false alarms (some examples are shown in Fig. 10) are caused by people waving flags, vigorously clapping hands, or sharply and disorderly waving hands.

By verifying the results, we can conclude that our proposed system is effective and robust for detecting violence with complex scenarios, such as various distances from cameras and severe occlusions between people and crowed scenes.

VI. CONCLUSION

This paper has targeted on effective STIP detection and feature representation for detecting violent behaviors in a real video scene. Following our recently proposed MoWLD, two major improvements have been made. First, considering the shortcomings of the well-known WLD, we have proposed the IWLD, and then proposed to extend the IWLD by adding a temporal component to the appearance descriptor to obtain MoIWLD. MoIWLD implicitly captures local motion information together with low-level image appearance information. Second, a modified sparse model has been proposed to learn a dictionary for classification. In the proposed sparse model, the representation-constrained term and the coefficient incoherence term have been introduced to ensure the learned dictionary to obtain a powerful discriminative ability. With this learned dictionary, both the representation residual and the representation coefficients are discriminative. Based on the proposed SRC, we have presented a corresponding classification scheme. The experimental results on three benchmark data sets have demonstrated the superiority of the proposed approach over the state-of-the-art approaches.

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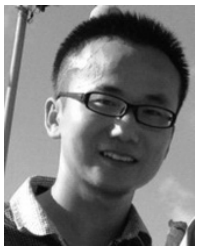

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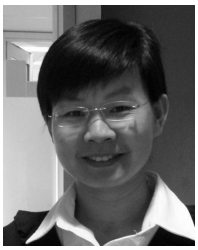

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